

# Automatic School Bell That Also Announces Alert Messages

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As the coronavirus pandemic continues, and could affect school-going children as well, the importance of smart devices has increased. This automatic school bell announces a recorded coronavirus alert after every school period. Due to

its repetition several times a day, children are likely to imbibe the message and start practicing the hygiene as per the message. The system is simple and uses low-cost devices.

## Circuit and working

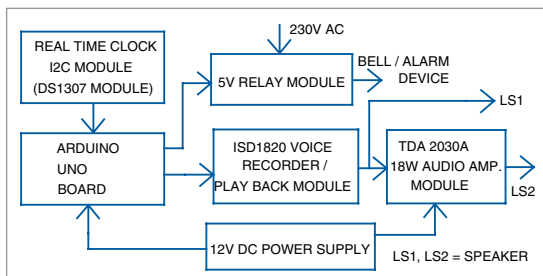


Fig. 1: Block diagram of the school bell with alert

Fig. 1 shows block diagram of the school bell that gives out a loud corona alert message. It comprises an Arduino Uno board, DS1307 real-time clock RTC module, TDA2030A 18W audio amplifier module, relay module, and speaker.

Fig. 2 shows circuit

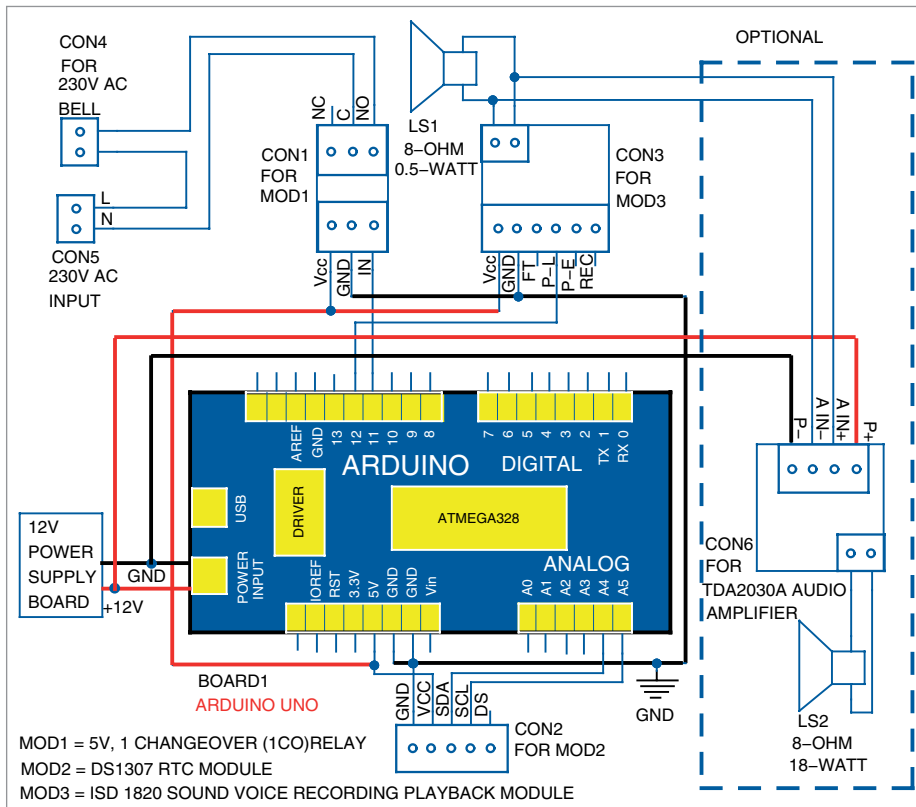


Fig. 2: Circuit diagram of the bell with alert

diagram of the school bell that works on 12V battery or adaptor. The project uses a voice recorder/playback module, which records the message and plays it back when triggered by the Arduino Uno board.

The voice record module shown in Fig. 3 is based on ISD1820 message record/playback device. It is a true single-chip voice recording device with a playback capability for 8 to 20 seconds. It has different modes of operation including REC, PLAYL, and PLAYE.

On pressing REC button RELED glows. The button has to be kept pressed until the recording ends and then released.

PLAYL button gives out a level activated signal. When its input pin level transits from low to high, a playback cycle is initiated. This mode can be activated by the pushbutton switch on the board, or by the microcontroller through a high-level pulse. Playback continues until PLAYL is pulled low, or an end-of-message (EOM) marker is detected, or the end of the memory space is reached.

The recorded playback is initiated by pulse produced by Arduino microcontroller as per program through output pin 12 of the Arduino board. The output power of ISD1820 voice recorder/playback is about 0.5 watts.

PLAYE button gives out a playback edge-activated signal. When this button is pressed, a high-going transition signal is detected on this pin and the device starts the playback cycle. Playback continues until an EOM marker is encountered, or the end of memory space is reached.

Fig. 4. shows TDA2030A 18W audio amplifier module that amplifies the voice output